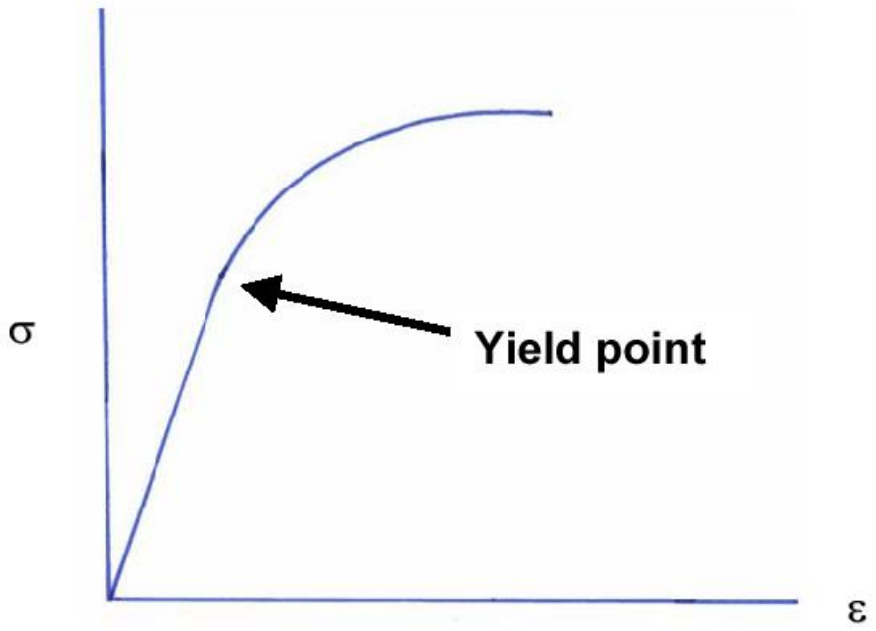
Outline

- Yield Strength
- Plastic Deformation
- Mechanical Behavior
- Safety Factor

Plastic Deformation

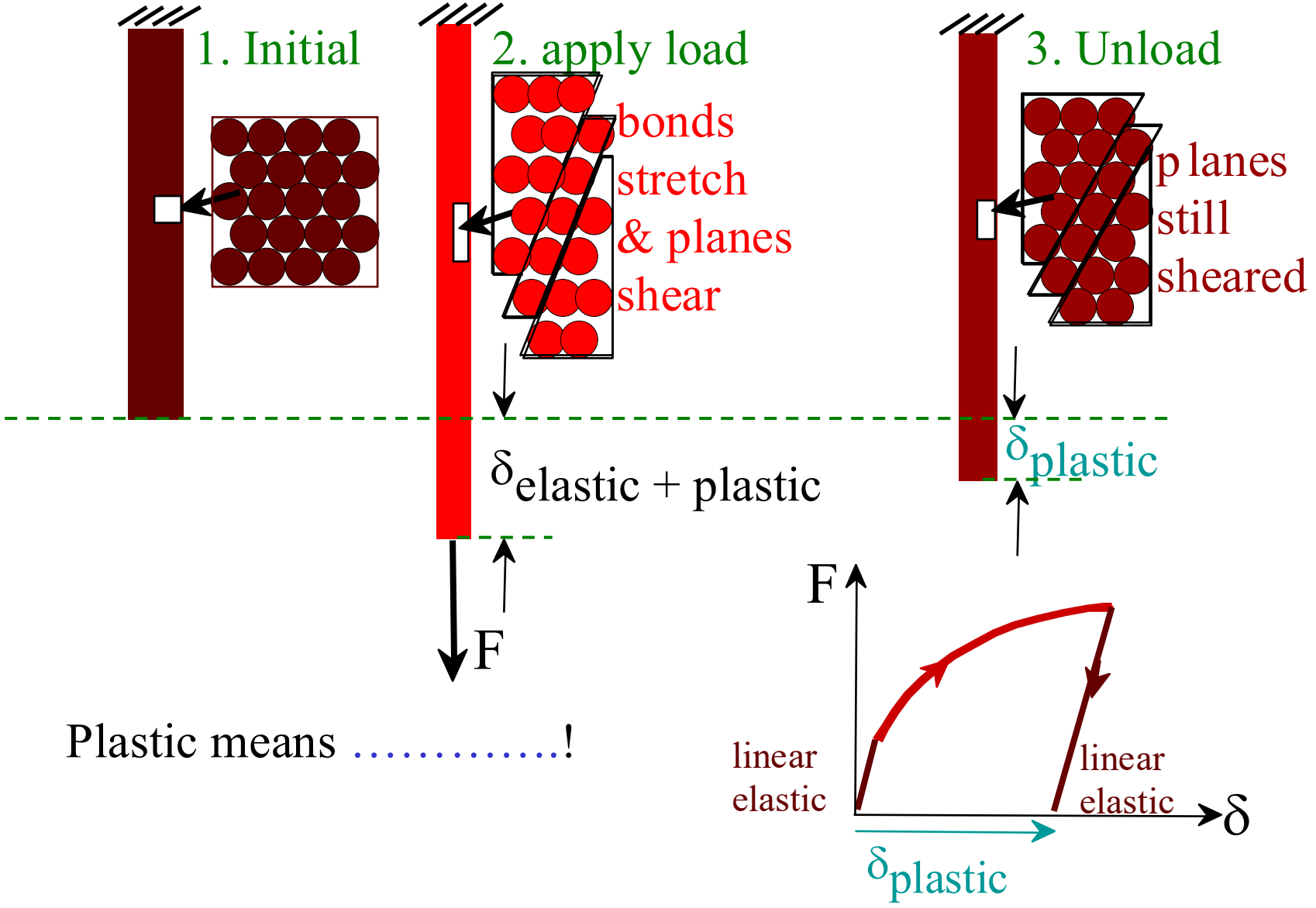
What happens if we continue to apply tensile loading beyond the elastic limit? *(i.e., stretching atomic bonds to the point of breaking)*



Plastic deformation:

- stress and strain are not proportional
- the deformation is not reversible
- deformation occurs by breaking and re-arrangement of atomic bonds (*in crystalline materials primarily by motion of dislocations, Chapter 7*)

Plastic Deformation: *metals*



Plastic means .....!

Proportional Limit:

in the text!

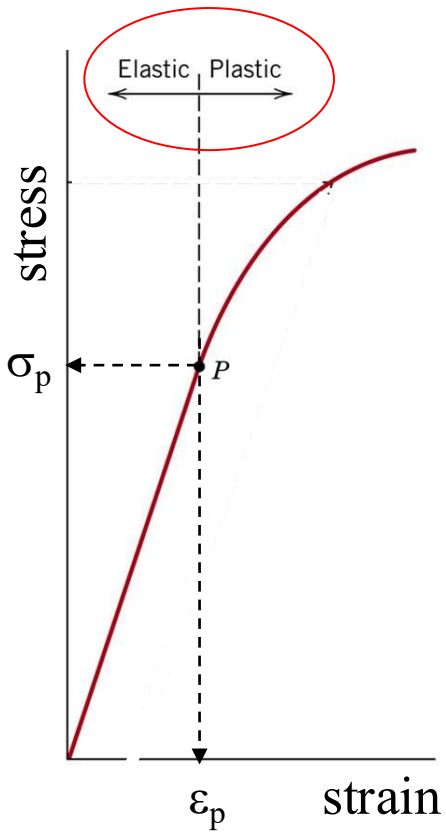
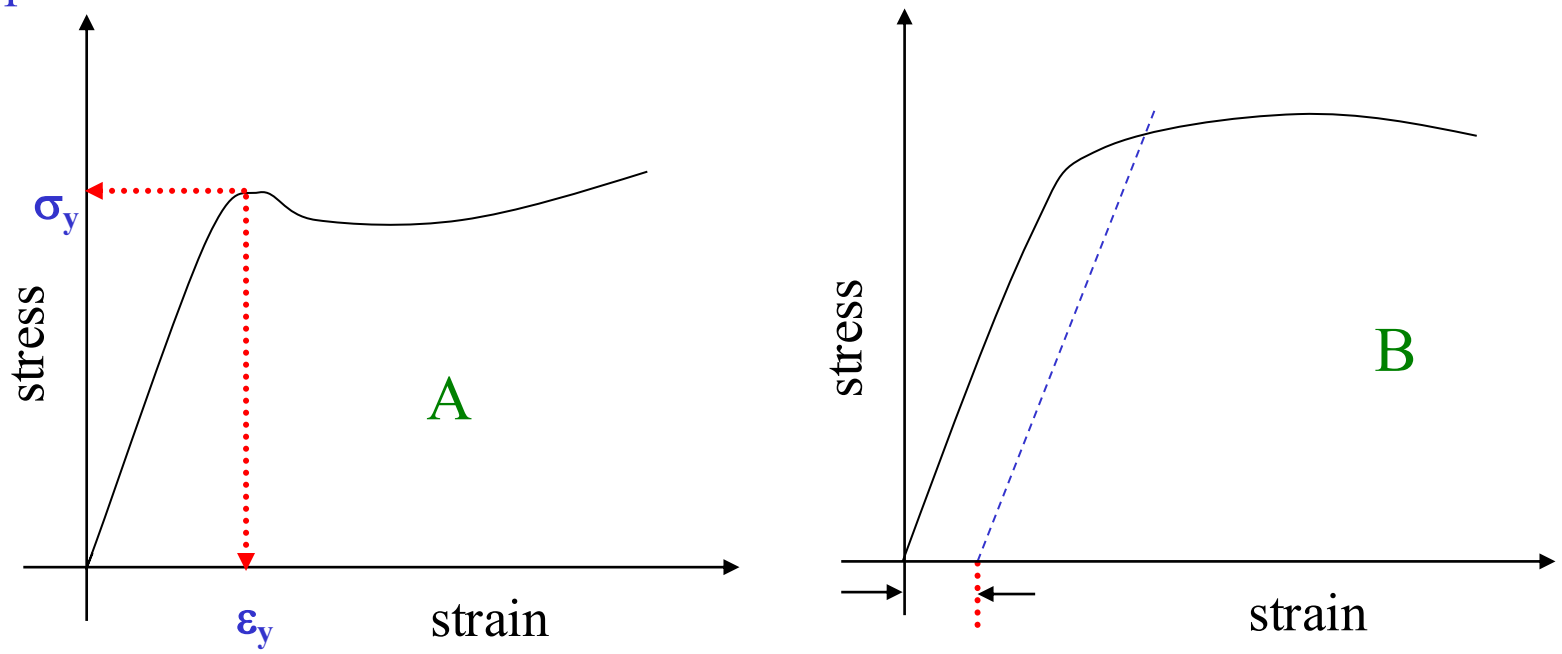


Figure 6.10 (a) – Callister 6th ed.

- Also known as the *elastic limit*, is the point where the stress-strain curve becomes nonlinear (the strain deviates from being proportional to the stress).
- The stress and strain values at this point are known as the proportional-limit stress and strain, respectively.
- This is the point beyond which **Hooke's law** can no longer be used to relate stress and strain in axial or shear deformation.

Yield Strength

• The yield point corresponds to the point where the material begins to have permanent deformation.

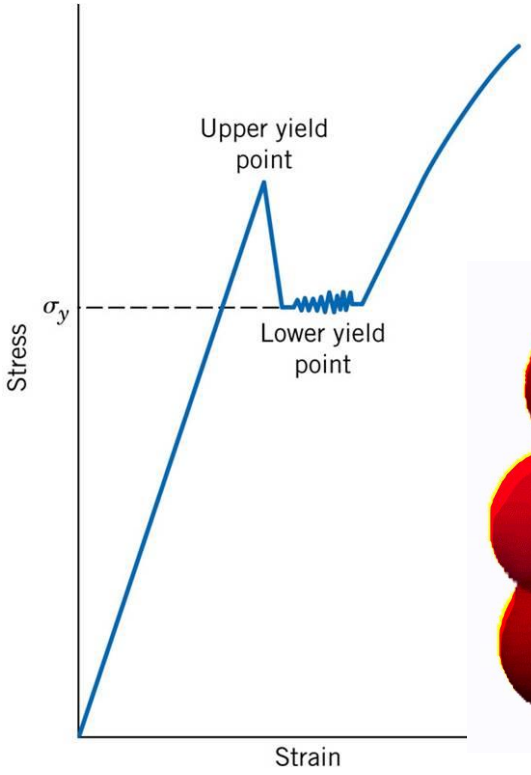


• Some materials have a well-defined yield region (A), others (B) do not.

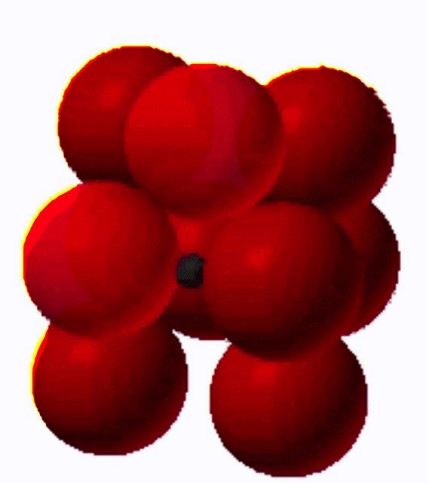
• In the absence of a distinct yield point, a 0.2% offset is used to obtain an approximate yield point.

• *Although the **yield** and the **proportional** limit points **are close** to each other, they **do not correspond** to the same location on the stress-strain curve.*

Yield Strength

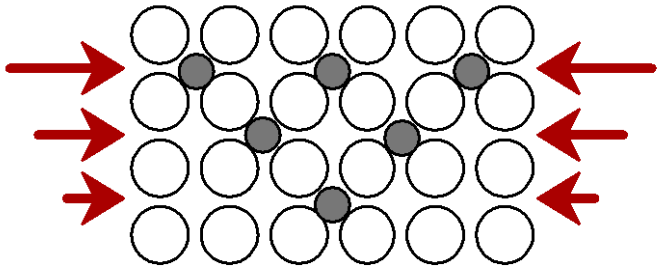


For a low-carbon steel, the stress vs. strain curve includes both an upper and lower yield point.



The yield strength is defined in this case as the average stress at the **lower yield** point.

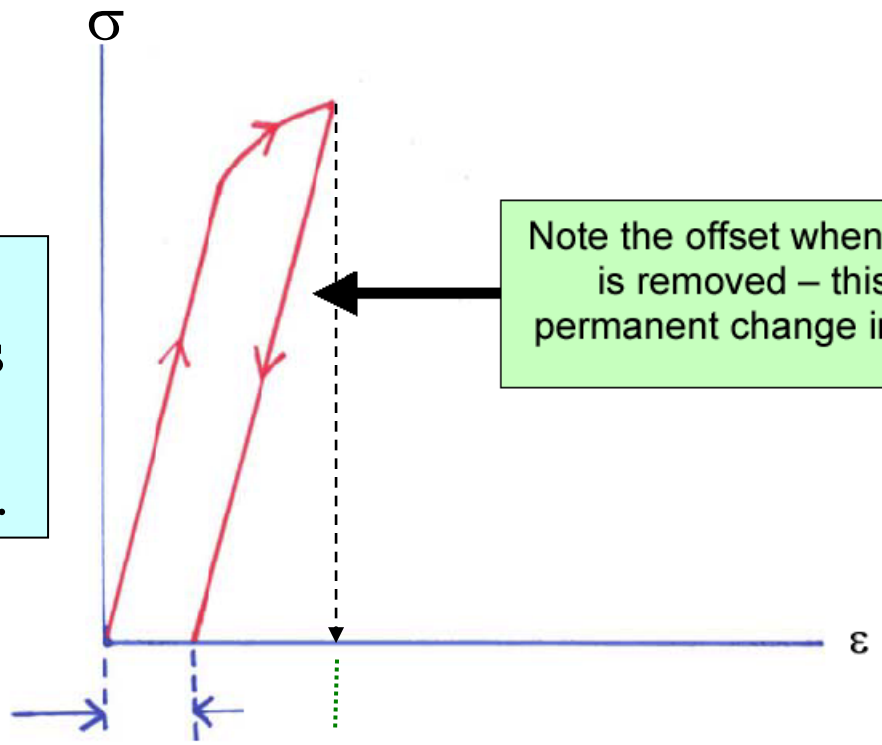
*Carbon and nitrogen atoms pin dislocations. The **upper yield point** marks their breakaway. Once free, dislocations move more easily, so plastic deformation continues at the **lower yield stress.***



Plastic Deformation

Suppose a tensile load is applied to a specimen and then released after the yield point was reached!

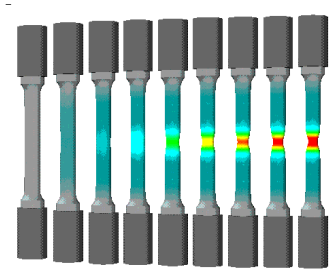
Plastic deformation is ……: when the stress is removed, the material does not return to its original dimension.

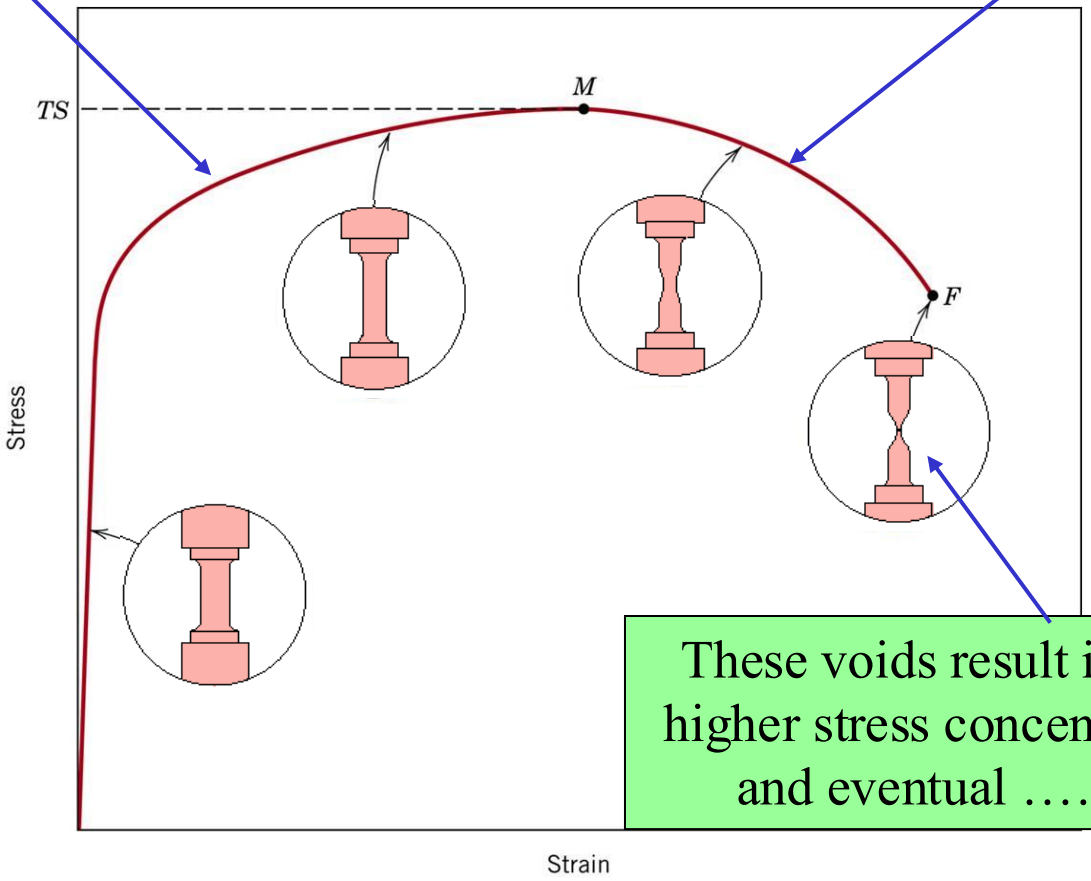


Mechanical Behaviour

As plastic deformation proceeds, the force increases due to …….

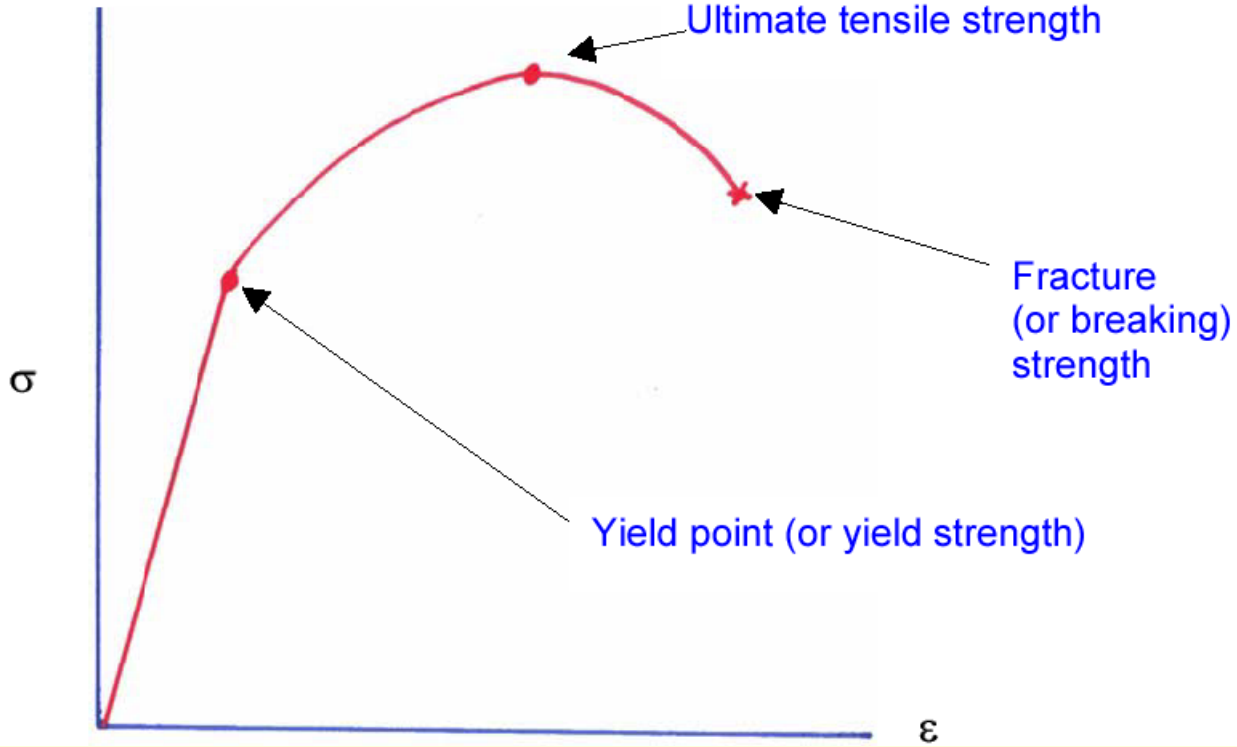
As more of the stress becomes concentrated in the neck, formation of ……. occur





These voids result in even higher stress concentrations and eventual ……

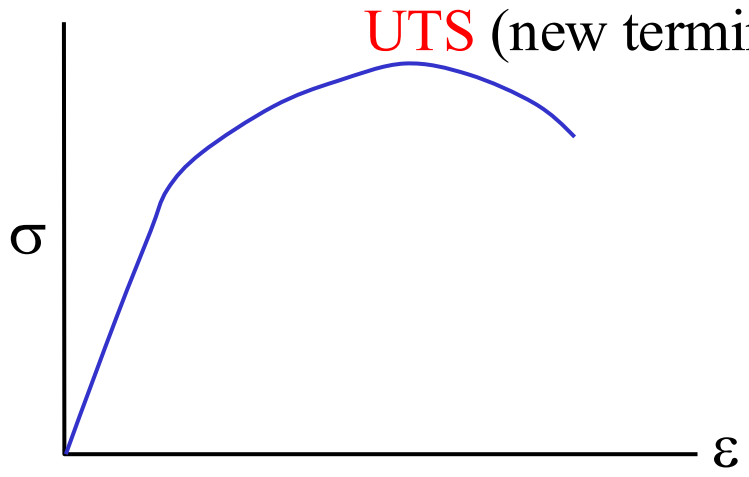
Mechanical Behavior



• For structural applications, the **yield stress** is usually a more ..... property than the **tensile strength**, since once it is passed, the structure has deformed beyond acceptable limits.

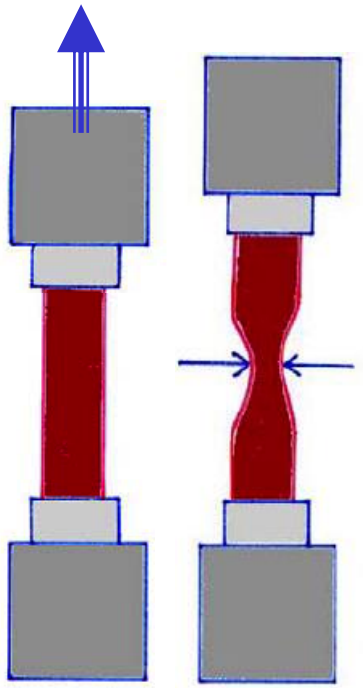
• However, parts produced by plastic deformation are worked between the yield and tensile strength, i.e. the TS point must not be passed.

Mechanical Behavior



Question: In ductile metals, the σ - ε curve eventually turns down after reaching the ultimate tensile strength (UTS). Does this mean the specimen is becoming "weaker"?

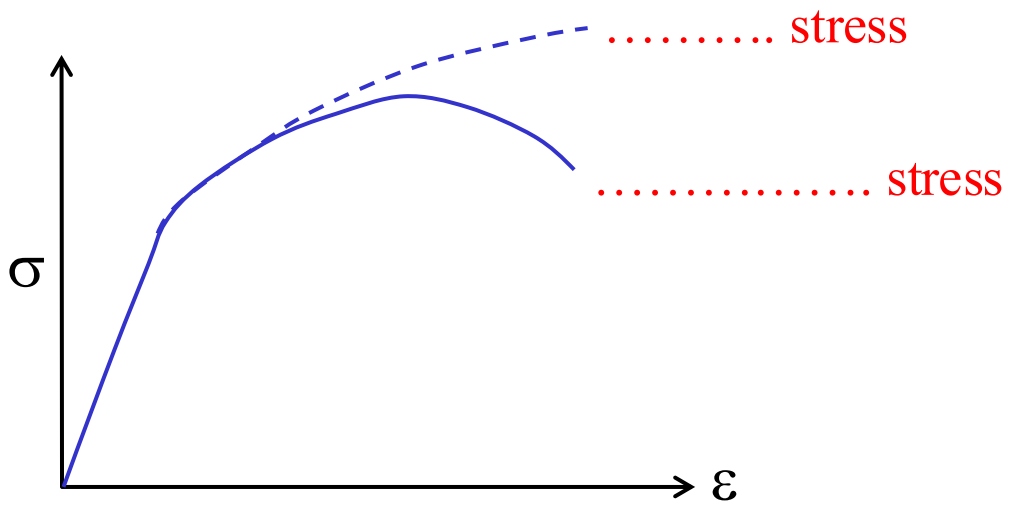
we know that the gauge area decreases during plastic deformation due to necking



Recall the definition of stress: "engineering" stress = F/A_o where A_o is the initial cross-sectional area

Engineering Stress vs. True Stress

Since the actual cross-sectional area is reduced, the use of the initial area gives a lower stress value than the actual one (the ratio is A_o/A_c).



- Even though the true stress-strain curve gives a more accurate picture of the breaking strength of a material, it is difficult to obtain measurements of *the actual area in real-time*.
- Usually, the reported values are for the engineering stress.
- True fracture strength > tensile strength
  - *but the engineering σ - ε diagram does not show this*

Example

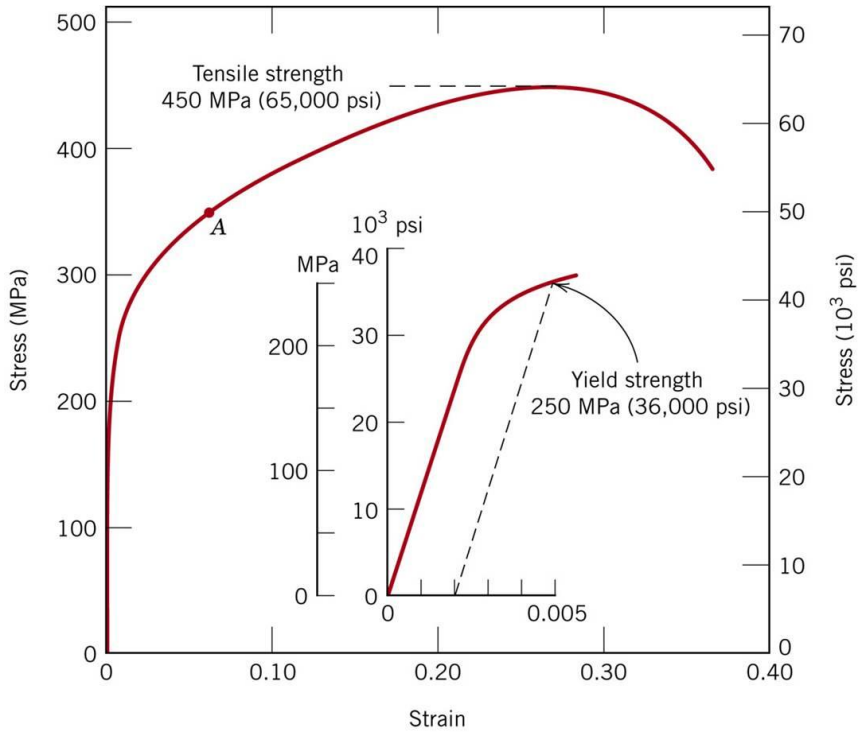
From the tensile σ - ε behaviour for a specimen of brass shown in the figure, determine the following:

a) modulus of elasticity

b) yield strength at a strain offset of 0.002

c) maximum load that can be sustained by a cylindrical specimen having an original diameter of 12.8 mm

d) change in length of a specimen originally 250 mm long that is subjected to a tensile stress of 345 MPa



a)

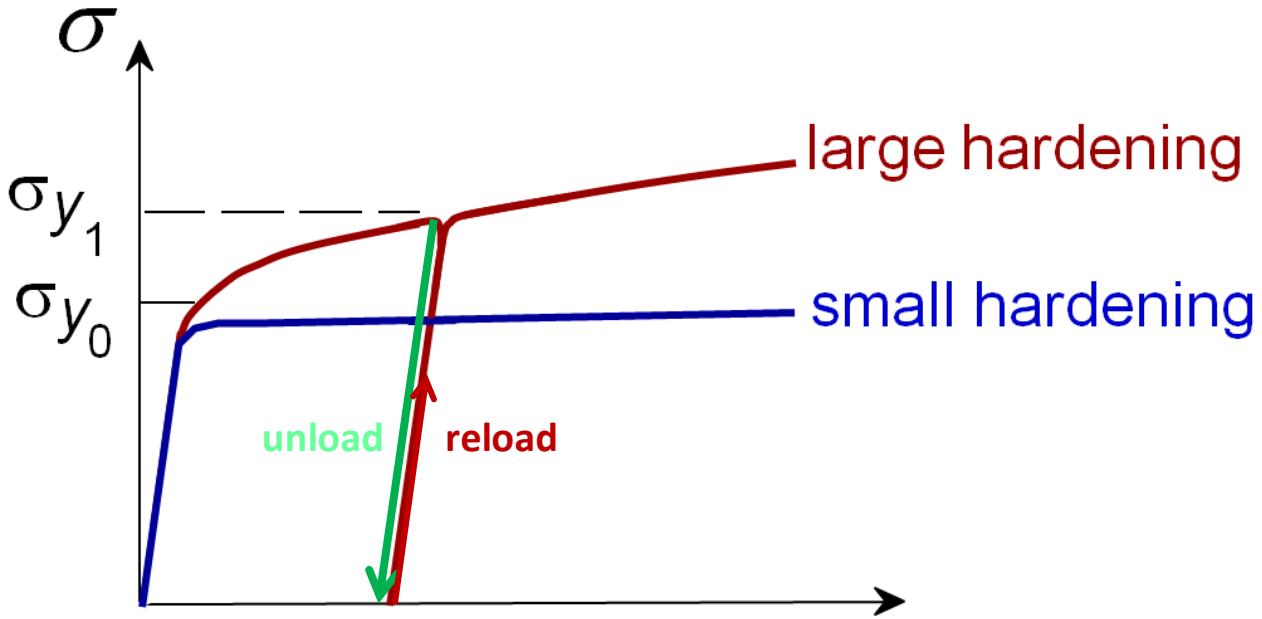
b)

c)

d)

Hardening

- An increase in σ_y due to plastic deformation.



- Curve fit to the stress-strain response:

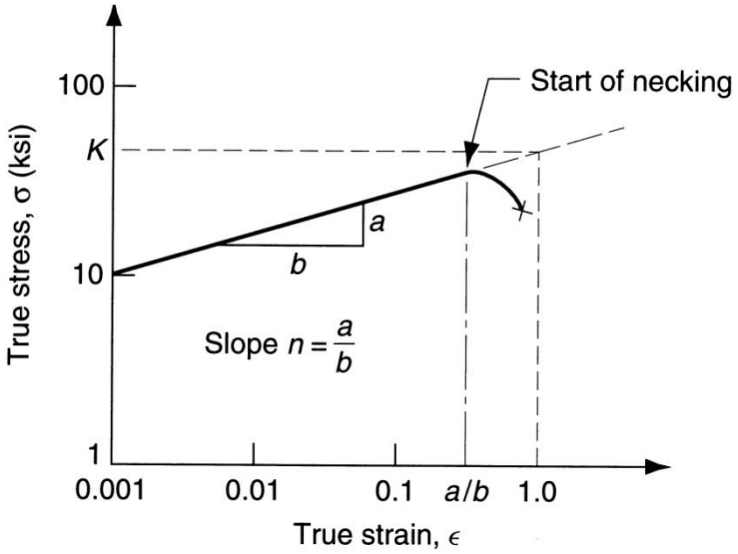
$$\sigma_T = K(\varepsilon_T)^n$$

hardening exponent:
n=0.15 (some steels)
to n=0.5 (some copper alloys)

"true" stress (F/A)

"true" strain: $\ln(L/L_o)$

True Stress-True Strain Curve



True stress-strain curve plotted on log-log scale

Material	n	K MPa	K psi
Low-carbon steel (annealed)	0.26	530	77,000
Alloy steel (Type 4340, annealed)	0.15	640	93,000
Stainless steel (Type 304, annealed)	0.45	1275	185,000
Aluminum (annealed)	0.20	180	26,000
Aluminum alloy (Type 2024, heat treated)	0.16	690	100,000
Copper (annealed)	0.54	315	46,000
Brass (70Cu–30Zn, annealed)	0.49	895	130,000

$\sigma = K\,\varepsilon^n$

Because it is a straight line in a log-log plot

K: strength coefficient

n: strain-hardening exponent

- The …… the slope the stronger when material is strained

Safety Factors in Design

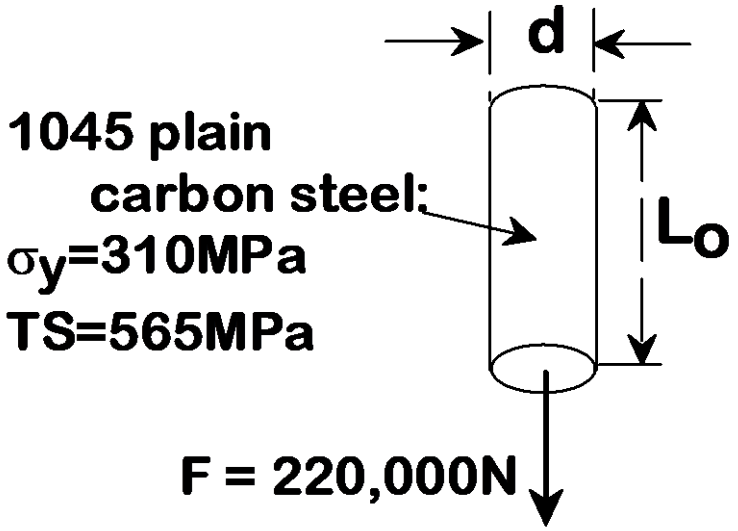
- Design uncertainties mean we must not push the limit.
- Factor of safety, N

$$\sigma_{working} = \frac{\sigma_y}{N}$$

Often N is between 1.2 and 4

Example: Calculate the diameter, *d*, to ensure that yielding does not occur in the 1045 carbon steel rod below. Use a factor of safety of 3.

$$\sigma_{working} = \frac{\sigma_y}{N}$$

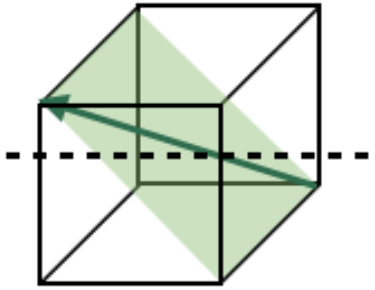


Review – Plastic Deformation

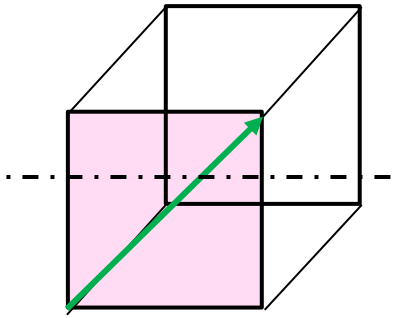
For the BCC crystal type which one of the following scenarios is more likely to exhibit slip?

Note: *the loading axis is depicted with a BLACK dotted line and is always normal to a cube face.*

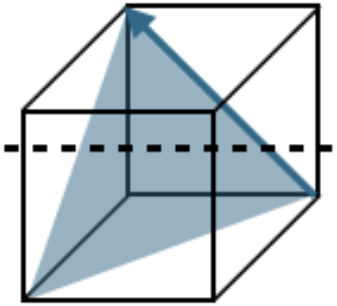
A



B



C



Use the QR code shown in the classroom. This one won't work.

Next time:

Toughness, Hardness…